Lösung: Fläche zwischen Graphen

a) Schnittpunkte:

$$f(x) = g(x) \quad \Rightarrow \quad x = 2 - x^2$$

$$\Leftrightarrow \quad x^2 + x - 2 = 0$$

$\Rightarrow$ PQ-Formel: $x_{1/2} = -\frac{1}{2} \pm \sqrt{\frac{1}{4} + 2} = -\frac{1}{2} \pm \sqrt{\frac{9}{4}} = \frac{-1 \pm 3}{2}$

$$\underline{\underline{x_1 = 1}} \qquad \underline{\underline{x_2 = -2}}$$

b) Fläche:

$$A = \left| \int_{-2}^{1} (f(x) - g(x))\, dx \right| = \left| \int_{-2}^{1} (2 - x^2 - x)\, dx \right|$$

$$= \left| \left[2x - \frac{x^3}{3} - \frac{x^2}{2} \right]_{-2}^{1} \right| = \left| 2 - \frac{1}{3} - \frac{1}{2} - \left(-4 + \frac{8}{3} - 2 \right) \right|$$

$$= \left| -\frac{9}{3} - \frac{1}{2} + 8 \right| = \left| \frac{-18 - 3 + 48}{6} \right| = \left| \frac{27}{6} \right| = |4{,}5| = \underline{\underline{4{,}5}}$$